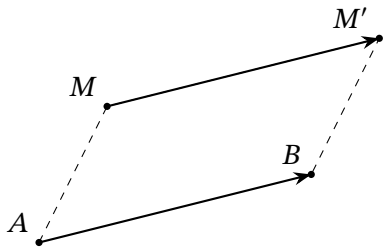
SEQUENCE 2

Vectors in the plane

14–30 September 2026

2.1 Translation and vectors

Key point · Definition. The translation that transforms A into B associates with every point M the unique point M' such that $ABM'M$ is a parallelogram, possibly flat. It is called the translation by the vector $\overrightarrow{AB}$. A nonzero vector has a direction, a sense and a norm. $\|\overrightarrow{AB}\| = AB$. The vector $\overrightarrow{AA}$ is the zero vector, written $\vec{0}$; it has neither direction nor sense.



Key point · Equal and opposite vectors. $\overrightarrow{AB} = \overrightarrow{CD}$ when the translation by the vector $\overrightarrow{AB}$ transforms C into D. Two nonzero vectors are equal if and only if they have the same direction, the same sense and the same norm. The opposite vector of $\overrightarrow{AB}$ is $\overrightarrow{BA}$: $\overrightarrow{BA} = -\overrightarrow{AB}$.

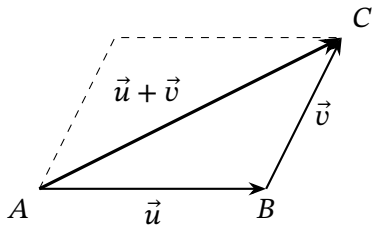
Example 1. Draw a segment $[AB]$ and place a point C outside (AB). Construct D such that $\overrightarrow{AB} = \overrightarrow{CD}$. Name the parallelogram obtained. Construct E such that $\overrightarrow{CE} = -\overrightarrow{AB}$.

2.2 Sum, difference and Chasles's relation

Key point · Sum of vectors. The sum $\vec{u} + \vec{v}$ corresponds to carrying out the translations by the vectors $\vec{u}$ and $\vec{v}$ one after the other. For all points A, B, C:

$$\boxed{\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}} \qquad \text{(Chasles's relation).}$$

We have $\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$ and $\overrightarrow{AB} + \overrightarrow{BA} = \vec{0}$.



Key point · Parallelogram rule. $ABDC$ is a parallelogram, possibly flat, if and only if $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{AC}$.

Example 2. Simplify: $\overrightarrow{AB} + \overrightarrow{BC}$; $\overrightarrow{AB} + \overrightarrow{CA}$; $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$; $\overrightarrow{AB} - \overrightarrow{AC}$.

Example 3. In a parallelogram $ABCD$, justify that $\overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AC}$.

2.3 Product of a vector by a real number

Key point · Definition and calculation rules. For $k \in \mathbb{R}$, $k\vec{u} = \vec{0}$ if $k = 0$ or $\vec{u} = \vec{0}$. Otherwise, $k\vec{u}$ has the same direction as $\vec{u}$, the same sense if $k > 0$, the opposite sense if $k < 0$, and its norm is $|k|\|\vec{u}\|$.

$$(k+\ell)\vec{u} = k\vec{u} + \ell\vec{u}, \quad k(\vec{u}+\vec{v}) = k\vec{u} + k\vec{v}, \quad k(\ell\vec{u}) = (k\ell)\vec{u}.$$

Example 4. For a vector $\vec{u}$ of norm 4 cm, represent $2\vec{u}$, $-\vec{u}$ and $\frac{1}{2}\vec{u}$ from the same origin. State their norms.

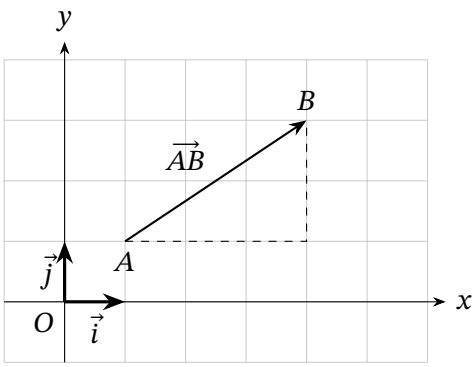
Key point · Characterisations of the midpoint. I is the midpoint of $[AB]$ if and only if $\overrightarrow{AI} = \overrightarrow{IB}$, or equivalently $\overrightarrow{AI} = \frac{1}{2}\overrightarrow{AB}$, or $\overrightarrow{IA} + \overrightarrow{IB} = \vec{0}$.

Example 5. In a triangle ABC, I is the midpoint of $[AB]$ and J that of $[AC]$. Show that $\vec{IJ} = \frac{1}{2}\overrightarrow{BC}$.

Fluency · 5–10 min. Simplify $3\vec{u} - 5\vec{u}$; calculate $(-3)^2$; solve $2x + 1 = 7$.

2.4 Frames, bases and coordinates

Key point · Definitions. Two non-collinear vectors $\vec{i}, \vec{j}$ form a basis of the plane. Every vector $\vec{u}$ can be written uniquely as $\vec{u} = x\vec{i} + y\vec{j}$; its coordinates are (x, y) or $\begin{pmatrix} x \\ y \end{pmatrix}$. A frame $(O, \vec{i}, \vec{j})$ associates with the point M the coordinates of $\overrightarrow{OM}$. It is **orthonormal** if the basis vectors are perpendicular and of norm 1.



Here $\overrightarrow{AB} = 3\vec{i} + 2\vec{j}$: we move three units to the right and two upwards.

Key point · Calculations with coordinates. If $A(x_A, y_A)$, $B(x_B, y_B)$, $\vec{u}(x, y)$ and $\vec{v}(x', y')$:

$$\overrightarrow{AB} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}, \quad \vec{u} + \vec{v} = \begin{pmatrix} x + x' \\ y + y' \end{pmatrix}, \quad k\vec{u} = \begin{pmatrix} kx \\ ky \end{pmatrix}.$$

Two vectors are equal if and only if their respective coordinates are equal.

Example 6. For $\vec{u}(3, 2)$ and $\vec{v}(-1, 4)$, calculate $\vec{u} + \vec{v}$ and $2\vec{u} - \vec{v}$. Write $\vec{w}(5, -3)$ in the basis $(\vec{i}, \vec{j})$.

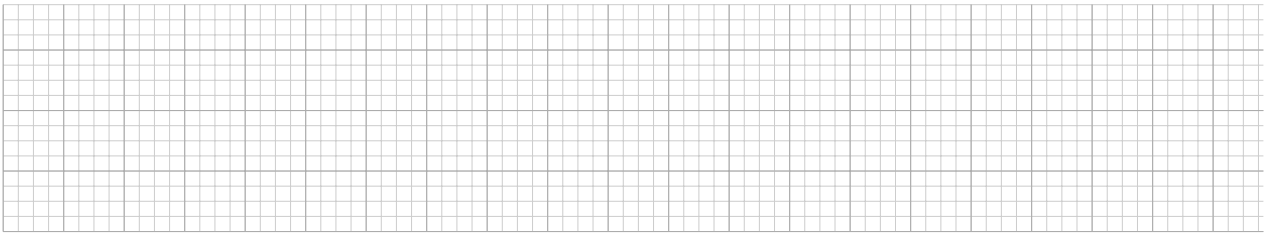
2.5 Norm, distance and midpoint

Key point · Properties. In an **orthonormal** frame, for $\vec{u}(x, y)$ and two points A, B:

$$\|\vec{u}\| = \sqrt{x^2 + y^2}, \qquad AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}.$$

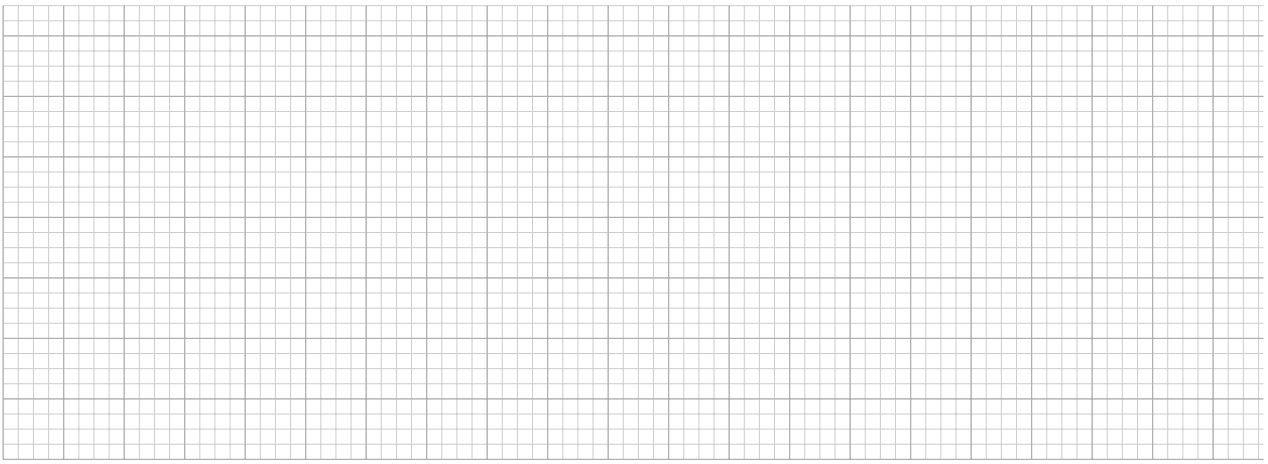
In any frame, the midpoint of $[AB]$ is $I\left(\frac{x_A+x_B}{2}; \frac{y_A+y_B}{2}\right)$.

Proof to write.



Example 7. In an orthonormal frame, $A(3, 0)$, $B(5, 3)$, $C(4, 0)$ and $D(6, 4)$.

1. Calculate $\overrightarrow{AB}$, $\overrightarrow{CD}$, then $3\overrightarrow{AB} - 5\overrightarrow{CD}$.
2. Determine U such that $\overrightarrow{BU} = 2\overrightarrow{AB} - \overrightarrow{CD}$.
3. Determine E such that $ABCE$ is a parallelogram, then its centre I.
4. Calculate AB and CD.



2.6 Collinearity and determinant

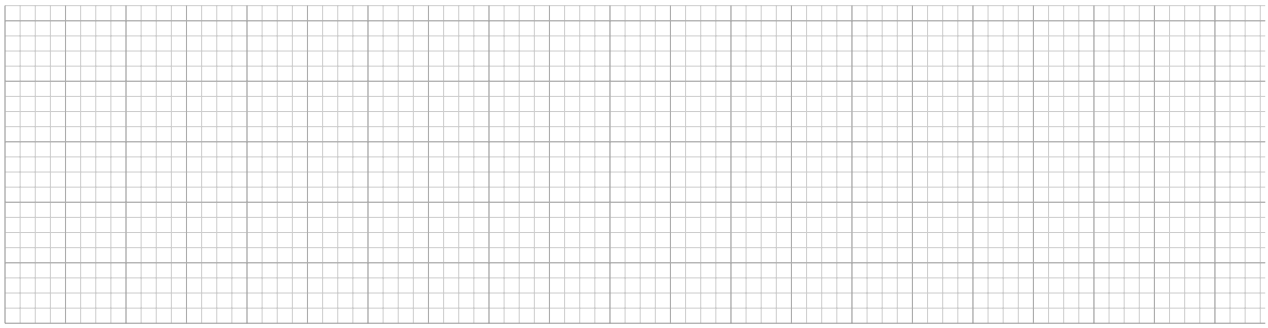
Key point · Definition. Two nonzero vectors are collinear if they have the same direction. The zero vector is collinear with every vector. If $\vec{v} \neq \vec{0}$, $\vec{u}$ and $\vec{v}$ are collinear if and only if there exists a real number k such that $\vec{u} = k\vec{v}$.

Watch out. In the last characterisation, the condition $\vec{v} \neq \vec{0}$ is necessary: a nonzero vector cannot be a multiple of the zero vector.

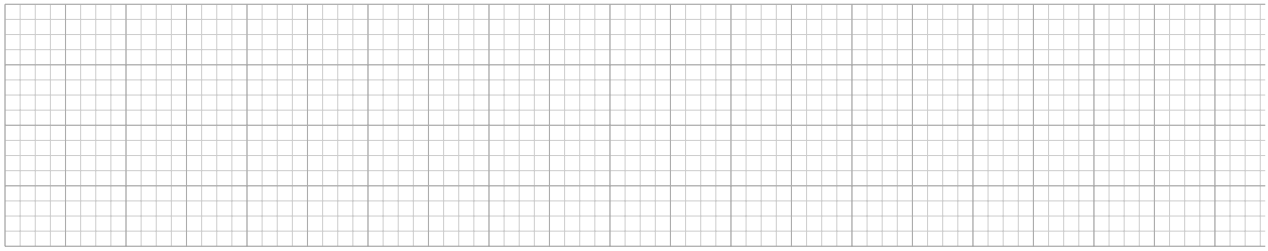
Key point · Determinant and collinearity criterion. For $\vec{u}(x, y)$, $\vec{v}(x', y')$, $\det(\vec{u}, \vec{v}) = xy' - yx'$.

$$\vec{u}, \vec{v} \text{ collinear} \iff \det(\vec{u}, \vec{v}) = 0.$$

Proof to write.



Example 8. Study the collinearity of: (a) $\vec{u}(3, 2)$ and $\vec{v}(-6, -5)$; (b) $\vec{w}(4, 1)$ and $\vec{t}(3, 3/4)$. If $\overrightarrow{AB} = 2\overrightarrow{CD}$ and $\overrightarrow{BD} = -5\overrightarrow{CD}$, relate $\overrightarrow{AB}$ and $\overrightarrow{BD}$.



2.7 Alignment and parallelism

Key point · Applications. For $A \neq B$ and $C \neq D$, the lines (AB) and (CD) are parallel (possibly coincident) if and only if $\det(\overrightarrow{AB}, \overrightarrow{CD}) = 0$. The points A, B, C are collinear if and only if $\det(\overrightarrow{AB}, \overrightarrow{AC}) = 0$.

Example 9. Show that (AB) and (CD) are parallel with $A(2, -4)$, $B(10, -2)$, $C(-16, -4)$ and $D(12, 3)$.

Example 10. Show that $E(0, 1/2)$, $F(2, 1)$ and $G(6, 2)$ are collinear.

Example 11. With $A(-1, 1)$, $B(3, 2)$, $C(-2, -3)$, $D(6, -1)$, $E(5, 0)$, show that $(AB) \parallel (CD)$ and that E, B, D are collinear.